

Dr. Chilperic Armel Foko Kuate

Computational Systems Biologist | Scientific Developer | Mechanistic Modeling

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★ RESEARCH AREAS & METHODS

- Systems biology and mechanistic modelling
- Hepatic metabolism and lipid dynamics
- Plant physiology and photosynthetic adaptation
- Quantitative immunology and T-cell dynamics
- ODE, DAE, and stochastic models
- Parameter estimation, calibration, and uncertainty analysis
- Constraint-based and differentiable metabolic modeling

</> PROGRAMMING & TOOLS

- </> Languages
Python, Julia, R, MATLAB
- ⚙️ Optimisation
CasADi, IPOPT, CMA-ES, JuMP, Pyomo
- 📊 Scientific Computing
SciPy, NumPy, CVODE, SUNDIALS, lmfit
- 📈 Data & Visualisation
pandas, matplotlib, ggplot2, Plotly, Bokeh
- 🔍 Sensitivity Analysis
SALib — Sobol, Morris
- 🏠 Applied ML
regression, classification; scikit-learn
- >_ Workflow
git, Jupyter, 🐧 Linux, HPC
- 🌐 Web Tools
HTML, CSS, JavaScript, browser-based interfaces, interactive simulation tools

LANGUAGES

- French Native
- English Fluent
- German B1+

🏆 AWARDS & FELLOWSHIPS

- ★ Bayer B4U Mentoring Programme, mentee (2026)
- ★ Marie Curie ITN Fellowship, PoLiMeR (2019–2023)
- ★ Post-AIMS Research Bursary (2017–2019)
- ★ F.K.A. Allotey Meritorious Award (2016)

👤 PROFILE

Computational systems biologist, applied mathematician, and scientific developer building mechanistic models and computational tools for biological systems, from hepatic metabolism and immune-cell dynamics to photosynthetic adaptation across the C3 to C4 continuum. My work combines dynamical systems theory, stochastic modeling, and biological interpretation, with a focus on what mathematics can explain about living systems, not just fit. I also develop scientific and educational web tools that make simulation, analysis, and technical training more accessible.

📁 RESEARCH EXPERIENCE

Independent Research, Scientific Software, and Further Training Nov 2025 – Present
Düsseldorf, Germany

- Preparing two manuscripts from postdoctoral research on the C3 to C4 continuum: a first-principles thermo-hydraulic framework for photosynthetic adaptation and a structure-function framework for leaf photosynthesis.
- Developed browser-based tools for scientific modeling and language learning, including Foko Lab and Deutsch-WiPA 2026.
- Pursuing structured upskilling in AI engineering and professional German to strengthen research-software development and scientific work in German-speaking environments.

Postdoctoral Researcher Nov 2022 – Oct 2025
CEPLAS / Institute of Computational Cell Biology, HHU Düsseldorf

- Developed a first-principles thermo-hydraulic-biochemical modeling framework for photosynthetic adaptation across the C3 to C4 continuum in *Flaveria*, linking mesophyll and bundle-sheath physiology to soil and environment.
- Coupled C3 to C4 FvCB biochemistry, heat balance, hydraulic regulation, and environmental forcing to explain how anatomy, transport, and physiology jointly shape photosynthetic performance.
- Applied Sobol sensitivity analysis and CMA-ES-based evolutionary simulations across contrasting climatic regimes to identify influential traits, quantify trade-offs, and show that different climates favor different adaptive routes rather than a single universal optimum.
- Presented this work at the CEPLAS Early Career Researcher Symposium, Düsseldorf, 2025.
- Supervised 2 MSc and 2 BSc projects in computational plant biology.

Marie Curie PhD Fellow Aug 2019 – Aug 2023
Institute of Quantitative and Theoretical Biology, HHU Düsseldorf

- Developed dynamic and stochastic models of hepatic fatty-acid metabolism to clarify regulatory mechanisms underlying metabolite dynamics and disease-relevant metabolic behavior.
- Built a coarse-grained model of fatty-acid metabolism and derived conditions for bi-stability, providing a mechanistic explanation for metabolic switching between fed and fasted states.
- Developed a semi-mechanistic model of fatty-acid de novo synthesis, fitted it to time-course data, and showed when production rates can be approximated by Michaelis–Menten laws, improving kinetic interpretability.
- Performed parameter estimation, sensitivity analysis, and uncertainty analysis for nonlinear biological systems to strengthen mechanistic interpretation of hepatic fatty-acid metabolism.
- Developed the enzyme-kinetic component of a differentiable constraint-based metabolic modeling framework, formalizing how kinetic constraints integrate into systems-level metabolic analysis; published in *Metabolic Engineering* (2022).

Research Fellow Nov 2017 – Mar 2019
AIMS Cameroon & AIMS Rwanda

- Modeled T-cell proliferation dynamics using stochastic and deterministic frameworks, with emphasis on branching processes, individual-based simulation, and moment-based analysis.
- Developed a stochastic master-equation framework for T-cell proliferation, derived mean and variance expressions for cell counts, and set up a maximum-likelihood fitting framework for CFSE-type data.

- ★ Best Student, AIMS Ghana (2016)
- ★ AIMS Ghana Hackathon Winner (2016)

EDUCATION

PhD, Quantitative & Theoretical Biology	
HHU Düsseldorf	2019–2023
Master 2 Mathematics & Applications	
Univ. Yaoundé I	2017–2019
MSc Mathematical Sciences	
AIMS Ghana	2015–2016
BSc Mathematics	
Univ. Douala	2009–2012

TEACHING & SUPERVISION

Supervised 2 MSc and 2 BSc projects in computational biology (HHU, 2019–2025).

Delivered 200+ teaching hours — mathematical modelling, ODEs, stochastic processes, Python (AIMS Rwanda, AIMS Cameroon, HHU).

SCIENTIFIC TOOLS

Foko Lab

Browser-based environment for ODE, optimisation, steady-state, and stochastic modeling workflows.

Open web tool

Deutsch-WiPA 2026

Interactive B1 to B2 German learning tool with conjugation practice, feedback loops, and error tracking.

Open web tool

REFERENCES

Prof. Dr. Martin J. Lercher
HHU Düsseldorf
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Prof. Dr. Oliver Ebenhöh
HHU Düsseldorf
oliver.ebenhoeh@hhu.de

Prof. Dr. Wilfred Ndifon
AIMS Global Network
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Prof. Dr. Barbara M. Bakker
University Medical Center Groningen
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SELECTED TRAINING & CONFERENCES

Bayer B4U Mentoring Programme (Mentee)	Feb 2026
Inaugural edition in Germany; one-to-one pairing with Bayer scientists.	
PoLiMeR Marie Curie ITN Doctoral Training	2020–2022
Training across metabolism, regulation, enzyme catalysis, and scientific communication.	
Advanced AI and Data Science Workshop	Dec 2021
HeiCAD, HHU Düsseldorf — neural networks, optimisation, and applied data science tools.	
AI Engineering Learning Track	2025–Present
Ongoing self-directed training in AI engineering, applied machine learning, and technical implementation workflows.	
Professional German Development	Ongoing
Continued German progression beyond B1+ toward stronger professional fluency in academic and technical settings.	
Clinic on Meaningful Modelling of Epidemiological Data	Jul 2018
AIMS South Africa and SACEMA — epidemiological modeling and data-driven inference.	

Talks

Oct 2025 CEPLAS Early Career Researcher Symposium, Düsseldorf, talk

Mar 2022 PoLiMeR Symposium, Düsseldorf, talk & poster

Jul 2019 Theoretical Biology Seminar, Univ. Rwanda

Aug 2018 Mathematics Against Malaria, Mitweida

Posters

Jul 2023 Economic Principles in Cell Biology Summer School

May 2022 Systems Biology of Mammalian Cells, Heidelberg

Jul 2020 International Systems Biology Group, Innsbruck

Jul 2018 MMED Epidemiological Modelling, Cape Town

SELECTED COMPUTATIONAL TOOLS

- Foko Lab:** browser-based environment for ODE simulation, optimisation, steady-state analysis, stochastic modelling, model examples, and export workflows.
- Photosynthesis climate adaptation model:** thermo-hydraulic-biochemical framework coupling leaf biochemistry, heat balance, hydraulics, soil/environment constraints, sensitivity analysis, and optimisation.
- Fatty-acid metabolism repositories:** reproducible ODE workflows for hepatic fatty-acid bi-stability and de novo synthesis.
- T-cell proliferation models:** stochastic and deterministic workflows for CFSE-type T-cell division data.

PUBLICATIONS

Published

- Foko Kuate C.A.**, Ebenhöh O., Bakker B.M., Raguin A. *Kinetic Rate Laws of the Enzymes Involved in Animal Fatty Acid Synthesis. Bioscience Reports*, 2023.
- Wilken S.E., Besançon M., Kratochvíl M., **Foko Kuate C.A.**, Trefois C., Gu W., Ebenhöh O. *Interrogating the Effect of Enzyme Kinetics on Metabolism Using Differentiable Constraint-Based Models. Metabolic Engineering*, 2022.
- Dourado H., **Foko Kuate C.A.**, Liebermeister W., Lercher M.J. *Growth Mechanics and the Emergence of Metabolic Oscillations in Growing Cells. bioRxiv*, 2025.

+2 manuscripts in preparation on plant adaptation across the C3–C4 continuum.